vibrational patterns, and that would explain why, even if they were copiously produced in the hot, early universe, few if any would have survived until today. Even if string theory is right, our only chance to see the superheavy vibrational patterns would be to produce them through high-energy collisions in particle accelerators. However, as current accelerators can reach only energies equivalent to roughly 1,000 times the mass of a proton, they are far too feeble to excite any but string theory's most placid vibrational patterns. Thus, string theory's prediction of a tower of particles with masses starting some million billion times greater than that achievable with today's technology is not in conflict with observations.

This explanation also makes clear that contact between string theory and particle physics will involve only the lowest-energy—the massless—string vibrations, since the others are way beyond what we can reach with today's technology. But what of the fact that most of the particles in Tables 12.1 and 12.2 are not massless? It's an important issue, but less troubling than it might at first appear. Since the Planck mass is huge, even the most massive particle known, the top-quark, weighs in at only .00000000000000000116 (about 10^{-17}) times the Planck mass. As for the electron, it weighs in at .00000000000000000000000034 (about 10^{-23}) times the Planck mass. So, to a first approximation—*valid to better than 1* part in 10^{17}—all the particles in Tables 12.1 and 12.2 do have masses equal to zero times the Planck mass (much as most earthlings' wealth, to a first approximation, is 0 times that of the Sultan of Brunei), just as "predicted" by string theory. Our goal is to better this approximation and show that string theory explains the tiny deviations from 0 times the Planck mass characteristic of the particles in Tables 12.1 and 12.2. But massless vibrational patterns are not as grossly at odds with the data as you might have initially thought.

This is encouraging, but detailed scrutiny reveals yet further challenges. Using the equations of superstring theory, physicists have listed every massless string vibrational pattern. One entry is the spin-2 graviton, and that's the great success which launched the whole subject; it ensures that gravity is a part of quantum string theory. But the calculations also show that there are *many* more massless spin-1 vibrational patterns than there are particles in Table 12.2, and there are